Summary in Graph

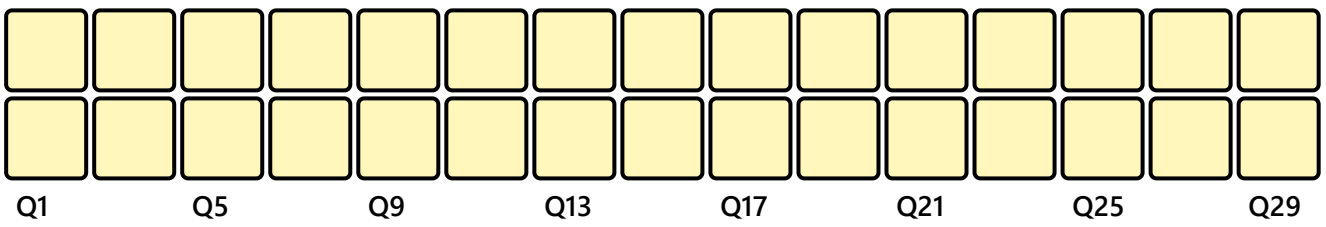
Exam Summary (GATE Overflow CS Test Series 2026 | Calculus | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (3) out of 32 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

If $f(x) = x^{100} + x^{99} + \cdots + x + 1$ then $f'(1)$ is equal to

A. 5050
B. 5049
C. 5051
D. none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

For which value of $'c'$ Lagrange's mean value theorem is applicable for the function $f(x) = \log x$ in $[1, e]$.

A. 1
B. e
C. $\frac{1}{e-1}$
D. $e - 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f : \mathbf{R} \to \mathbf{R}$ be defined by $f(x) = 2x + \cos x$. f :

A. has a minimum at $x = \pi$
B. has a maximum at $x = 0$
C. is a decreasing function
D. is an increasing function

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The maximum value of the real valued function $\sin x + \cos x$ is

A. 2
B. 1
C. 0
D. $\sqrt{2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

For which positive integer n is

$$\lim_{x \to 3} \frac{x^n - 3^n}{x - 3} = 108?$$

A. 3
B. 4
C. 5
D. 6

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

The value of $\lim_{x \to 0} \frac{\sqrt{2+x} - \sqrt{2}}{x}$.

A. $\frac{1}{\sqrt{2}}$
B. $\frac{1}{2\sqrt{2}}$
C. 0
D. Limit does not exist

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #7 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

The value of $\lim_{x \to 0} \frac{\sin(2+x) - \sin(2-x)}{x}$ is

A. $2\cos 2$
B. 1
C. $2\sin 2$
D. does not exist

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #8 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

The value of limit

$$\lim_{x \to \frac{\pi}{6}} \frac{2\sin^2 x + \sin x - 1}{2\sin^2 x - 3\sin x + 1}$$

A. 3
B. -3
C. 1
D. limit does not exist

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

$\lim_{x \to 0} \frac{\sin x}{x(1+\cos x)}$ is equals to

A. 0
B. $\frac{1}{2}$
C. 1
D. -1

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #10 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

$\lim_{x \to 0} \frac{|x|}{x}$ is equals to

A. 1
B. -1
C. 0
D. limit does not exist

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The value of

$$\lim_{x \to 0} x \sin \frac{1}{x}$$

A. 1
B. 0
C. $\frac{1}{2}$
D. Does not exist

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f(x)$ be the function defined as

$$f(x) = \begin{cases} \frac{k \cos x}{2x - \pi} & \text{when } x \neq \frac{\pi}{2} \\ 3 & \text{when } x = \frac{\pi}{2} \end{cases}$$

If $\lim_{x \to \frac{\pi}{2}} f(x) = f\left(\frac{\pi}{2}\right)$, the value of k is

A. 6
B. -6
C. 8
D. -8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let f be the function defined as

$$f(x) = \begin{cases} x + 2 & \text{when } x \leq -1 \\ cx^2 & \text{when } x > -1 \end{cases}$$

Which of the following is the correct value of $'c'$ if $\lim_{x \to -1} f(x)$ exists?

A. 1
B. -1
C. 0

D. none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f(x) = \begin{cases} \frac{\sin[x]}{[x]} & \text{when } [x] \neq 0 \\ 0 & \text{when } [x] = 0 \end{cases}$, where [.] denotes the greatest integer function, then $\lim_{x \to 0} f(x)$ is equal to

A. 1
B. 0
C. -1
D. none of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f(x) = \begin{cases} x^2 - 1; & 0 < x < 2 \\ 2x + 3; & 2 \leq x < 3 \end{cases}$, the quadratic equation whose roots are $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$ is

A. $x^2 - 6x + 9 = 0$
B. $x^2 - 7x + 8 = 0$
C. $x^2 - 14x + 49 = 0$
D. $x^2 - 10x + 21 = 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Numerical Type | Award: 2 | Penalty: 0 | Calculus

If $\lim_{x \to 0} \sin(mx) \cot\left(\frac{x}{\sqrt{3}}\right) = 2$ then $m =$ _____ (rounded off up to 2 decimal places)

Your Answer: | Correct Answer: 1.15 : 1.16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f(x) = \frac{1}{4x^2 + 2x + 1}$ then its maximum value is

A. 0.75
B. 1.33
C. 0.25
D. It does not have a maximum value

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The minimum value of the function

$$f(x) = \frac{x^3}{3} - x$$

occurs at

A. $x = 1$
B. $x = -1$
C. $x = 0$
D. $x = \frac{1}{\sqrt{3}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f(x)$ be the function defined as

$$f(x) = \begin{cases} |x| & \text{for } 0 < x \leq 2 \\ 1 & \text{for } x = 0 \end{cases}$$

At $x = 0$, f has:

A. has local maximum
B. has local minimum
C. no local maximum
D. no extremum

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Calculus

The maximum value of $f(x) = 2x^3 - 9x^2 + 12x - 3$ in the interval $0 \leq x \leq 3$ is _____

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

While minimizing a continuous function $f(x)$, necessary and sufficient conditions for a point x_0 to be a minima are

A. $f'(x_0) > 0$ and $f''(x_0) = 0$
B. $f'(x_0) < 0$ and $f''(x_0) = 0$
C. $f'(x_0) = 0$ and $f''(x_0) < 0$
D. $f'(x_0) = 0$ and $f''(x_0) > 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Calculus

The minimum value of the function $f(x) = \frac{1}{3}x(x^2 - 3)$ in the interval $-100 \leq x \leq 100$ occurs at $x =$ _____

Your Answer: | Correct Answer: -100 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Which of the following function is strictly decreasing?

A. $f(x) = ax + b$ for $x \in \mathbb{R}, a, b > 0$
B. $f(x) = e^{2x}$ for $x \in \mathbb{R}$
C. $f(x) = \frac{2}{x} + 5$ for $x \in \mathbb{R}$
D. $f(x) = x^3 - 6x^2 + 15x - 18$ for $x \in \mathbb{R}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Find the 'c' of Rolle's mean value theorem for $f(x) = x^3 - 4x \in [-2, 2]$.

A. $\frac{2}{\sqrt{3}}$
B. $-\frac{2}{\sqrt{3}}$
C. $\pm\frac{2}{\sqrt{3}}$
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Compute the value of

$$\int_0^{\frac{\pi}{2}} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx$$

A. π
B. $\frac{\pi}{4}$
C. 1
D. $\frac{\pi}{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following function(s) is/are continuous at $x = 3$? (Mark all the appropriate choices)

A. $f(x) = \begin{cases} 2; & \text{when } x = 3 \\ x - 1; & \text{when } x > 3 \\ \frac{x+3}{3}; & \text{when } x < 3 \end{cases}$

B. $f(x) = \begin{cases} 4; & \text{when } x = 3 \\ 8 - x; & \text{when } x \neq 3 \end{cases}$

C. $f(x) = \begin{cases} x + 3; & \text{when } x \leq 3 \\ x - 4; & \text{when } x > 3 \end{cases}$

D. $f(x) = \begin{cases} \frac{1}{\sqrt{x^3 - 27}}; & \text{when } x < 3 \end{cases}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the following definite integral

$$I = \int_0^1 \frac{(\sin^{-1} x)^2}{\sqrt{1 - x^2}} dx$$

The value of the integral is

A. $\frac{\pi^3}{24}$

B. $\frac{\pi^3}{12}$

C. $\frac{\pi^3}{48}$

D. $\frac{\pi^3}{64}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

What is the value of the integration

$$I = \int_0^{\frac{\pi}{2}} \log(\sin x)\, dx$$

A. 0

B. $\frac{-\pi}{2} \log 2$

C. $-\pi \log 2$

D. $\log 2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29

Numerical Type | Award: 2 | Penalty: 0 | Calculus

The value of the integral

$$I = \int_{-1}^{2} (1 + |x|)\, dx$$

$=$ _____ (rounded up to 1 decimal place)

Your Answer: | Correct Answer: 5.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The value of the integral

$$I = \int_{-\infty}^{\infty} \frac{dx}{1+x^2}$$

A. $-\pi$
B. $-\frac{\pi}{2}$
C. $\frac{\pi}{2}$
D. π

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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